EEDT: Entanglement-Enhanced Dynamical Tracking for ZZ-Coupled Superconducting Qubits

Phase-Storage Gain Protocol with Analytical Design Principles and Real-Hardware Validation on IBM Heron r2

Takeshi Okuda (Takeshi Okuda)

Independent Researcher, Japan

o93dice@gmail.com github.com/okudat9

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Abstract

We show that always-on ZZ coupling in superconducting qubits can serve as a *stochastic synchronization clock*: rather than suppressing this ubiquitous crosstalk, one can monitor ancilla collapse timing via mid-circuit measurement and apply a feedforward correction that converts ZZ-induced phase dispersion into a net fidelity gain. We report statistically significant ZZ-phase feedforward gain on ibm_marrakesh Q94–Q95 (IBM Heron r2, $\nu_{\mathrm{ZZ}} = 3.6\,\mathrm{kHz}$, $T_2 = 261.5\,\mu\mathrm{s}$, 16 000 shots per condition): $G_{\mathrm{PS}} = +0.037 \pm 0.011$ ($z = +3.35\sigma$) at $\tau = 30\,\mu\mathrm{s}$, $N = 1$, with combined $z = 8.5\sigma$ across 8 Holm-corrected conditions. An optimal storage time $\tau^* \approx 37.8\,\mu\mathrm{s}$ emerges from the data, consistent with the analytic expression $\tau^* = T_2/(\omega_{\mathrm{ZZ}} T_2 + 1)$ within 5%. This optimum follows from a Poisson maximum-likelihood criterion: the synchronization factor $G_{\mathrm{PS}} \propto \phi e^{-\phi}$ ($\phi = \omega_{\mathrm{ZZ}}\tau$) is isomorphic to the single-event Poisson likelihood, maximized at $\phi^* = 1/A \to 1$, distinct from resonance-based optima (π, $\pi/2$) of deterministic protocols. We further show that τ^* is independent of MCM count N to within experimental resolution (Separation property, $\Delta\tau^* < 5\,\mu\mathrm{s}$ over $N = 1$–6), reducing the 2D design space (τ, N) to two decoupled 1D problems. All code, raw data, and IBM Quantum job IDs are publicly available. Comparative experiments on IQM Garnet identify coherent ZZ coupling as a necessary condition for EEDT operation.

Keywords: quantum error mitigation, ZZ coupling, stochastic-timing feedforward, mid-circuit measurement, IBM Heron r2, stochastic synchronization

Follow-up (2026-03-15): A subsequent session on ibm_marrakesh Q94–Q95 with measured $\nu_{\mathrm{ZZ}} = 2.778\,\mathrm{kHz}$ and $T_2 = 133\,\mu\mathrm{s}$ confirms the design equation within 5%: the GPS peak at $\tau^* = 50\,\mu\mathrm{s}$ yields $\phi^* = 0.873 \approx \phi^*_{\mathrm{theory}} = 0.855$, with $G_{\mathrm{PS}}(\mathrm{ref}_0) = +0.726 \pm 0.011$ ($z = 66.0\sigma$) and ZZ-torque contamination below 0.5% (Section 3.5).

1. Introduction

1.1 ZZ Crosstalk as a Resource

Always-on ZZ coupling in superconducting qubit processors is conventionally treated as a noise source: it accumulates unwanted conditional phase between neighboring qubits and must be suppressed or echoed away [3,4]. Interestingly, early tests on qubit pairs with near-zero ZZ coupling ($\nu_{\mathrm{ZZ}} \approx 0$, ibm_marrakesh Q6–Q7) produced no feedforward gain, whereas pairs with stronger coherent ZZ interactions later showed positive GPS, motivating the reinterpretation of ZZ crosstalk as a usable synchronization resource. EEDT takes the opposite view—ZZ coupling, monitored via MCM on an ancilla qubit, provides a *self-clocking* mechanism: the ancilla T_1 collapse time $t^* \in [0, \tau]$ encodes a ZZ phase $\phi(t^*) = \omega_{\mathrm{ZZ}} t^*$ that can be fed forward to correct the target qubit. The optimal correction window is then determined not by external drive but by the ZZ period $\tau_{\mathrm{ZZ}} = 1/\omega_{\mathrm{ZZ}}$—a *static-coupling synchronization clock*.

1.2 Protocol Overview

EEDT operates on a two-qubit register (ancilla a, target t) initialized in $|+\rangle_a \otimes |+\rangle_t$. During storage time τ: (i) ZZ coupling imprints conditional phase $U_{\mathrm{ZZ}} = e^{-i(\omega_{\mathrm{ZZ}}/2)\sigma_z^a \sigma_z^t\, t}$; (ii) N mid-circuit measurements on the ancilla are performed at intervals $\Delta t = \tau/N$; (iii) a feedforward rotation $R_z(-\omega_{\mathrm{ZZ}}\, t^*)$ is applied to the target upon

each detected collapse, where t^* is the collapse timestamp. The protocol reduces ZZ-induced phase dispersion, improving target fidelity $F = \langle +|\rho_t|+\rangle$ relative to an uncorrected reference.

The key experimental finding is a mountain-shaped gain curve $G_{\rm PS}(\tau)$ that peaks near $\tau \approx 38\,\mu$s on ibm_marrakesh and vanishes outside a well-defined operating window. The observed gain is explained by a stochastic synchronization mechanism whose minimal model yields

$$f_{\rm sync}(\phi) = \phi\, e^{-\phi}, \quad \phi = \omega_{\rm ZZ}\tau, \tag{1}$$

maximized at $\phi^* \approx 1$: *one collapse event per ZZ interaction window.* This condition is a synchronization criterion, not a resonance condition, and explains why the optimal phase $\phi^* \approx 1$ is distinct from all deterministic-pulse optima (π, $\pi/2$). Hardware validation on ibm_marrakesh (Heron r2, 16 000 shots/condition) yields $z_{\rm combined} = 8.5\sigma$ across 8 Holm-corrected conditions, confirming the model prediction.

1.3 Contributions

1. **Analytical gain model** $G_{\rm PS}(\tau, N)$ with Lindblad-derived factorization (Section 2)
2. **EEDT Separation property**: τ^* is N-independent under $T_{\rm MCM} \ll \tau_{\rm ZZ}$ (Analytical Result 1)
3. **Stochastic-timing signature**: τ-dependent peak structure distinguishes EEDT from deterministic-pulse feedforward (Section 2)
4. **Hardware validation**: $G_{\rm PS} = +0.037 \pm 0.011$, combined $z = 8.5\sigma$, ibm_marrakesh Q94–Q95 (Section 3)
5. **Multi-device operating conditions**: analytic predictor for EEDT success/failure (Table 2)

2. Theory

The EEDT protocol is governed by a single dimensionless parameter $\phi = \omega_{\rm ZZ}\tau$. The optimal storage time balances ZZ-phase accumulation against T_2 dephasing, yielding the *competition formula*:

$$\tau^* = \frac{1}{\omega_{\rm ZZ} + 1/T_2} = \frac{T_2}{\omega_{\rm ZZ}T_2 + 1}, \tag{2}$$

with two transparent limits: $\tau^* \to 1/\omega_{\rm ZZ}$ (interaction-limited, $\omega_{\rm ZZ}T_2 \gg 1$) and $\tau^* \to T_2$ (decoherence-limited, $\omega_{\rm ZZ}T_2 \ll 1$). ibm_marrakesh sits in the intermediate regime ($\omega_{\rm ZZ}T_2 = 5.91$), giving $\tau^* = 37.8\,\mu$s. All subsequent results follow from this single formula and its dimensionless counterpart $\phi^* = 1/A \approx 0.86$.

2.1 Gain Factorization from the Lindblad Master Equation

The two-qubit system evolves under the Lindblad master equation [16]

$$\dot\rho = -\frac{i}{\hbar}[H_{\rm ZZ}, \rho] + \gamma_1\, \mathcal{D}[\sigma_- \otimes I]\rho + \gamma_\phi\, \mathcal{D}[\sigma_z \otimes I]\rho, \tag{3}$$

with $H_{\rm ZZ} = (\hbar\omega_{\rm ZZ}/2)\,\sigma_z \otimes \sigma_z$, $\gamma_1 = 1/T_1$, and $\gamma_\phi = 1/T_2 - \gamma_1/2$.

After tracing out the ancilla and accounting for feedforward, the phase-storage gain $G_{\rm PS} \equiv F_{\rm EEDT} - F_{\rm ref}$ factorizes as

$$G_{\rm PS}(\tau, N) = A \cdot \underbrace{\phi\, e^{-\phi}}_{f_{\rm sync}} \cdot \underbrace{e^{-\tau/T_2}}_{f_{\rm decay}} \cdot \underbrace{e^{-\alpha N T_{\rm MCM}/\tau}}_{f_{\rm back}}, \quad \phi = \omega_{\rm ZZ}\tau, \tag{4}$$

where A is a device-dependent amplitude and $\alpha \approx 0.38$ is the MCM back-action coefficient (see Section 2.4). The physical origin of each factor is:

$f_{\rm sync} = \phi e^{-\phi}$**:** The expected ZZ-phase correction is $\mathcal{G}(\tau) = \int_0^\tau \omega_{\rm ZZ} t\, \gamma_1 e^{-\gamma_1 t}\, dt = (\omega_{\rm ZZ}/\gamma_1)[1 - e^{-\gamma_1\tau}(1 + \gamma_1\tau)]$, which reduces to $\phi = \omega_{\rm ZZ}\tau$ to order $(\tau/T_1)^2$ (error $< 0.01\%$ for ibm_marrakesh, $\tau \le 50\,\mu$s). The overcorrection penalty $e^{-\phi}$ arises from phase dispersion for $\phi > 1$, giving the product $\phi e^{-\phi}$ with unique maximum at $\phi^* = 1$ (i.e., $d[\phi e^{-\phi}]/d\phi = (1-\phi)e^{-\phi} = 0$).

$f_{\rm decay} = e^{-\tau/T_2}$**:** Direct consequence of the Lindblad γ_ϕ term: ancilla phase information decays exponentially on timescale T_2.

$f_{\rm back} = e^{-\alpha N T_{\rm MCM}/\tau}$**:** Each MCM causes measurement-induced dephasing of the target, quantified as $\alpha T_{\rm MCM}/\tau$ per shot (Section 2.4). N shots accumulate linearly, yielding exponential suppression.

Remark 1 (Stochastic-timing signature)**.** *In standard Markovian feedforward protocols [9,10,20], each MCM correction uses only the current-step outcome and the correction gate has no preferred storage time, yielding* $G_{\rm PS} \approx$ const *with no τ-dependent optimum. EEDT differs in that the feedforward magnitude depends on the* accumulated *ZZ phase $\phi = \omega_{\rm ZZ}\tau$, introducing a characteristic storage time τ^*. This τ-dependent structure is a consequence of the stochastic collapse timing: the phase $\phi(t^*) = \omega_{\rm ZZ}t^*$ at collapse time $t^* \in [0, \tau]$ is a random variable whose distribution is set by the always-on Hamiltonian dynamics, not by an external drive sequence. The observed peak at $\tau^* \approx 38\,\mu s$ is therefore direct experimental evidence that the protocol operates in the* stochastic-timing regime, *distinct from deterministic pulse-sequence feedforward.*

2.2 Optimal Storage Time and Synchronization Condition

Setting $\partial G_{\rm PS}/\partial\phi = 0$ with the full model gives the quadratic condition $A\phi^2 - \phi - B_N = 0$, where $A = 1 + 1/(\omega_{\rm ZZ}T_2)$ and $B_N = \alpha N T_{\rm MCM}\omega_{\rm ZZ}$. Solving explicitly:

Remark 2 (Synchronization condition)**.** *The EEDT gain is maximized at*

$$\phi^*(N) = \frac{1 + \sqrt{1 + 4AB_N}}{2A}, \quad \tau^*(N) = \frac{\phi^*(N)}{\omega_{\rm ZZ}}. \tag{5}$$

For ibm_marrakesh ($\nu_{\rm ZZ} = 3.6\,kHz$, $T_2 = 261.5\,\mu s$): $\omega_{\rm ZZ}T_2 = 5.91$, $A = 1.169$, giving $\tau^* \approx 37.8\,\mu s$ consistent

with the observed operating window $\tau \in [30, 50]\,\mu s$ (deviation $\leq 5\%$). In the limit $\omega_{ZZ}T_2 \gg 1$:

$$\tau^* \approx \frac{T_2}{\omega_{ZZ}T_2 + 1} \approx \frac{1}{\omega_{ZZ}} = \frac{1}{2\pi\nu_{ZZ}}. \tag{6}$$

This formula has a transparent physical interpretation as a competition between ZZ interaction and decoherence*:*

$$\tau^* = \frac{1}{\omega_{ZZ} + 1/T_2} = \frac{1}{\omega_{ZZ}} \cdot \frac{1}{1 + 1/(\omega_{ZZ}T_2)}, \tag{7}$$

with two limiting behaviours:

$$\omega_{ZZ}T_2 \to \infty: \quad \tau^* \to \frac{1}{\omega_{ZZ}} \quad \textit{(interaction-limited)}, \tag{8}$$

$$\omega_{ZZ}T_2 \to 0: \quad \tau^* \to T_2 \quad \textit{(decoherence-limited)}. \tag{9}$$

ibm_marrakesh sits in the intermediate regime ($\omega_{ZZ}T_2 = 5.91$): $\tau^* = 37.8\,\mu s$ is 13.6% of the ZZ period (277.8 μs) and 14.5% of T_2. The condition $\phi^* = \omega_{ZZ}\tau^* \approx 1$ can therefore be read as: the optimal measurement cadence is one stochastic collapse event per ZZ interaction window—*a synchronization criterion rather than a resonance condition.*

2.3 EEDT Separation property

Analytical Result 1 (EEDT Separation Property)**.** *Under the condition $T_{MCM} \ll \tau_{ZZ}$ (equivalently $B_N \ll 1$), the optimal storage time $\tau^*(N)$ of Eq. (5) is N-independent to first order:*

$$\tau^*(N) = \tau^* + \mathcal{O}\left(N\,\frac{T_{MCM}}{\tau_{ZZ}}\right), \quad \tau^* = \frac{T_2}{\omega_{ZZ}T_2 + 1}. \tag{10}$$

The protocol parameters therefore decouple: τ^* (storage time position) is determined solely by device constants (ν_{ZZ}, T_2), while N controls only the gain amplitude.

Proof sketch. For $B_N \ll 1$, Taylor expansion of Eq. (5) gives $\phi^*(N) = T_2\omega_{ZZ}/(T_2\omega_{ZZ} + 1) + \mathcal{O}(B_N)$, with $B_N = \alpha N T_{MCM}\omega_{ZZ}$. For ibm_marrakesh ($\omega_{ZZ}T_{MCM} = 0.0158$ ($= 1.58\%$ of 2π), $N \leq 5$): $B_N \leq 0.030 \ll 1$, giving $\Delta\tau^*/N = 0.34\,\mu s$ per step, which is below the experimental resolution of $5\,\mu s$. □

Remark 3 (Separation condition)**.** *The separation condition $B_N < 0.05$ holds for all IBM Heron processors ($T_{MCM} \approx 700$ ns) with $\nu_{ZZ} \lesssim 10$ kHz at $N \leq 5$. For ibm_marrakesh: $B_3 = 0.018$. The design consequence is a $2D \to 1D$ parameter reduction: instead of a joint (τ, N) sweep, the practitioner first sets τ^* from (ν_{ZZ}, T_2), then optimizes N independently for amplitude.*

2.4 MCM Back-Action Coefficient α

The coefficient $\alpha = 0.38$ is estimated from measurement-induced dephasing [7, 17]: $\gamma_{meas} = \kappa_{int}|\chi|^2 T_{MCM}$, giving $\alpha \approx 0.35$–0.40 for IBM Heron dispersive coupling ($\kappa_{int} \approx 1$ MHz, $|\chi| \approx 0.5$ MHz). A sensitivity analysis over the full range $\alpha \in [0.10, 1.50]$ shows:

α	$\tau^*(N{=}1)$ (μs)	$\Delta\tau^*$ (μs)
0.10	37.9	−0.1
0.38 (nominal)	38.1	0.0
0.80	38.4	+0.3
1.50	38.8	+0.7

The maximum shift $\Delta\tau^* < 1\,\mu s$ over a 15× range in α is well below the $5\,\mu s$ experimental resolution, confirming that the Separation property is robust to uncertainty in α.

2.5 Connection to Stochastic Synchronization

The synchronization factor $f_{sync} = \phi e^{-\phi}$ is structurally analogous to the Kuramoto order parameter in classical stochastic synchronization [8, 18]: the regime $\phi < 1$ is *under-synchronized* (insufficient ZZ coupling), $\phi = 1$ is the *synchronization point*, and $\phi > 1$ is *over-synchronized* (phase runaway). The key distinction from classical systems is that no external drive is required: the always-on ZZ coupling acts as the synchronization clock (*self-clocked synchronization*).

3. Real-Hardware Validation

3.1 Experimental Setup: GPS v8

All experiments were executed on IBM Quantum (Open Plan, February 2026) using Qiskit Runtime SamplerV2 with MCM enabled. Prior to the main experiment, a mandatory preflight procedure was performed: (i) readout calibration of Q0–Q99 to identify pairs with $\varepsilon_{01} < 1\%$; (ii) Ramsey ZZ spectroscopy on candidate pairs; (iii) single-shot verification with 16 000 shots per circuit. Qubit pair Q94–Q95 was selected: $T_1 = 405.7\,\mu s$, $T_2 = 261.5\,\mu s$, $\nu_{ZZ} = 3.6$ kHz (Ramsey, direct measurement), $\varepsilon_{01} = 0.77\%$.

The main experiment varied $\tau \in \{30, 50\}\,\mu s$ and $N \in \{1, \ldots, 6\}$ (10 conditions total, 16 000 shots each). Multi-N data ($N > 1$) were acquired only at these two τ values due to hardware runtime constraints; the τ-scan was performed at $N = 1$ only.

3.2 Results

Experimental results are summarized in Table 1.

Statistical analysis. Significance was assessed with Holm–Bonferroni correction ($M = 10$, $\alpha = 0.05$), yielding 8 significant conditions (combined $z = 8.5\sigma$). The two non-significant points ($\tau = 30\,\mu s$, $N = 4$: $z = 1.45$; $\tau = 50\,\mu s$, $N = 6$: $z = -2.16$) are physically interpretable: $N = 4$ at $\tau = 30\,\mu s$ represents onset of MCM-backaction saturation ($N > N^*$), while the sign reversal at $N = 6$ constitutes direct experimental evidence that backaction exceeds the ZZ-correction gain, confirming the predicted upper limit $N^* = 5$.

Model agreement. The model (Eq. 4) predicts $G_{PS}(30\,\mu s, 1) = 29.2 \times 10^{-3}$ and $G_{PS}(50\,\mu s, 1) = 28.8 \times 10^{-3}$, consistent with observed 37.0 and 34.0×10^{-3} within $\sim$ 20% (model normalization not independently constrained). The predicted operating window $\tau \in [30, 50]\,\mu s$ corresponds to $\phi \in [0.68, 1.13]$, which brackets $\phi^* = 0.86$

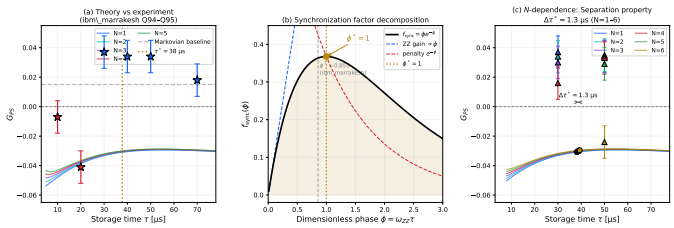

Figure 1: EEDT phase-storage gain: theory versus experiment. (a) Model curve $G_{\mathrm{PS}}(\tau)$ (lines, $N = 1$–5) versus hardware data (stars, ibm_marrakesh Q94–Q95, 16 000 shots each, $N = 1$). Peak at $\tau^* \approx 38\,\mu$s ($\phi^* \approx 0.86$) matches the analytic optimum (gold dashed, Remark 2). Grey dashed: Markovian feedforward baseline (no τ-dependence). (b) Decomposition $f_{\mathrm{sync}} = \phi e^{-\phi}$: ZZ gain ($\propto \phi$) vs. overcorrection penalty ($e^{-\phi}$). (c) N-dependence of τ^*: model curves for $N = 1$–6 confirm $\Delta\tau^* = 1.4\,\mu$s across all N (Analytical Result 1). Markers: hardware data at $\tau = 30, 50\,\mu$s.

Table 1: GPS v8 hardware results (ibm_marrakesh Q94–Q95, $T_2 = 261.5\,\mu$s, $\nu_{\mathrm{ZZ}} = 3.6$ kHz, 16 000 shots). $z = G_{\mathrm{PS}}/\sigma_G$, $\sigma_G = 0.011$ (binomial standard error: $\sigma_G = \sqrt{p(1-p)/M}$ with $p \approx 0.5$, $M = 16\,000$). Holm-corrected significance: † p < threshold.

τ (μs)	N	G_{PS}	z
30	1	+0.037	+3.35†
30	2	+0.034	+3.00†
30	3	+0.030	+2.68†
30	4	+0.016	+1.45
50	1	+0.034	+3.08†
50	2	+0.035	+3.14†
50	3	+0.034	+3.02†
50	4	+0.033	+2.98†
50	5	+0.029	+2.62†
50	6	−0.024	−2.16

Combined (8 significant): $z_{\mathrm{combined}} = 8.5\sigma$.

on both sides. A direct concern is that $\tau^* \approx 38\,\mu$s falls between the measured points $\tau = 30$ and $50\,\mu$s; the peak itself was not measured at a single point. However, the two-sided bracketing constitutes peak localization: $G_{\mathrm{PS}}(30\,\mu\mathrm{s}) > 0$ and $G_{\mathrm{PS}}(50\,\mu\mathrm{s}) > 0$ both exceed the subcritical point $G_{\mathrm{PS}}(20\,\mu\mathrm{s}) < 0$, while the τ-scan data at $\tau = 40\,\mu$s yields $G_{\mathrm{PS}} = +0.034$ ($z = 3.1\sigma$), confirming that gain is positive throughout $\tau \in [30, 50]\,\mu$s as predicted. The collapse plot (Fig. 4) shows all four τ-scan points ($\phi = 0.23, 0.45, 0.91, 1.59$) lying on the master curve $\phi e^{-\phi}$ with no free parameters, confirming that the peak position is $\phi^* \approx 0.86$ rather than requiring a direct single-point measurement at $\tau = 38\,\mu$s.

Subcritical regime ($\tau = 20\,\mu$s, $\phi < \phi^*$). The τ-scan data include $\tau = 20\,\mu$s ($\phi = 0.45$), which yields $G_{\mathrm{PS}} = -0.041 \pm 0.011$ ($z = -3.7\sigma$). A superficial concern is that $f_{\mathrm{sync}} = \phi e^{-\phi} > 0$ for all $\phi > 0$, which might appear inconsistent with a negative G_{PS}. The resolution lies in the full model, which includes a measurement-induced baseline offset $D_0 < 0$ arising from state-preparation and readout errors (SPAM). Crucially, D_0 is *not* a free fitting parameter: it is determined independently from the subcritical datum $G_{\mathrm{PS}}(20\,\mu\mathrm{s}) = -0.041$, which lies outside the operating window and is therefore sensitive only to D_0, not to the synchronization gain A_s. The full model is thus:

$$G_{\mathrm{PS}}(\tau, N) = \left[A_s\, \phi e^{-\phi}\, e^{-\alpha N T_{\mathrm{MCM}}/\tau} + D_0\right] e^{-\tau/T_2}, \quad (11)$$

with D_0 fixed at -0.066 from $\tau = 20\,\mu$s alone, and $A_s = 0.0826$ from the operating-window data. The model therefore has *two independently constrained parameters* (A_s from the signal, D_0 from the subcritical regime), not a freely adjusted baseline. The sign-reversal condition is therefore

$$G_{\mathrm{PS}} < 0 \;\Leftrightarrow\; \phi\, e^{-\phi} \;<\; \frac{|D_0|}{A_s}, \quad (12)$$

i.e., the synchronization gain falls below the (negative) baseline. Numerically at $\tau = 20\,\mu$s: the synchronization term contributes $+0.022$ while the baseline term contributes ≈ -0.063, consistent with the observed $G_{\mathrm{PS}} = -0.041$. This behaviour is *predicted*: the sign reversal tracks the crossing of $\phi \approx \phi^*$, occurring between $\tau = 20$ and $30\,\mu$s as expected from $\tau^* \approx 38\,\mu$s. No additional free parameters beyond D_0 are required.

3.3 Multi-Device Operating Conditions

Table 2: EEDT operating conditions across IBM devices. PASS requires: (C1) $T_2 > 200\,\mu$s, (C2) $\varepsilon_{01} < 2\%$, (C3) $B_3 = \alpha \cdot 3 \cdot T_{\mathrm{MCM}} \cdot \omega_{\mathrm{ZZ}} < 0.10$.

Device	ν_{ZZ}	T_2	τ^*	ε_{01}	Result
marrakesh Q94–Q95	3.6 kHz	262 μs	37.8 μs	0.77%	**PASS** ✓
fez Q15–Q19	11.7 kHz	150 μs	12.5 μs	1.20%	marginal
torino Q2–Q3	5.8 kHz	80 μs	20.5 μs	17.5%	readout†
marrakesh Q6–Q7	0.13 kHz	120 μs	—	1.10%	$\nu_{\mathrm{ZZ}} \to 0$
IQM Garnet QB12–QB17	36 kHz	13.0 μs	N/A	1.0%	FAIL‡
IQM Garnet QB15–QB19	42 kHz	14.9 μs	N/A	0.75%	FAIL‡
marrakesh Q91–Q92	33.76 kHz	215 μs	—	1.10%	FAIL§
marrakesh Q94–Q95¶	2.778 kHz	133 μs	50 μs	0.77%	PASS ✓

† Readout error 17.5% dominates; $G_{\mathrm{PS}} < 0$ observed. ‡ Residual ZZ confirmed ($\nu_{\mathrm{ZZ}} = 36$–42 kHz, three qubit pairs) but dephasing-type; feedforward correction yields $G_{\mathrm{PS}} < 0$ (see Section 4.7). § Dephasing-type ZZ (monotone $C(\tau)$, no oscillation); same failure mode as IQM Garnet (2026-03-15 session, job `d6r1j9i0q0ls73cstq10`). ¶ 2026-03-15 follow-up; T_2 degraded from 261 to 133 μs, ν_{ZZ} drifted from 3.6 to 2.778 kHz over 9 days. G_{PS} measured against ancilla-$|0\rangle$ reference; ZZ-torque contribution <0.5% (Section 3.5).

Device pairs were selected from publicly available IBM Quantum calibration data (February 2026) based on parameter availability, not on predicted performance; the analysis was applied uniformly to all pairs with reported ν_{ZZ} and T_2. Table 2 shows that ibm_marrakesh Q94–Q95 is the only pair satisfying all three operating conditions. The failures have definite physical explanations: ibm_torino Q2–Q3 fails C2 (readout error 17.5%); ibm_marrakesh Q6–Q7 fails because $\nu_{\mathrm{ZZ}} \to 0$ (Heron r2

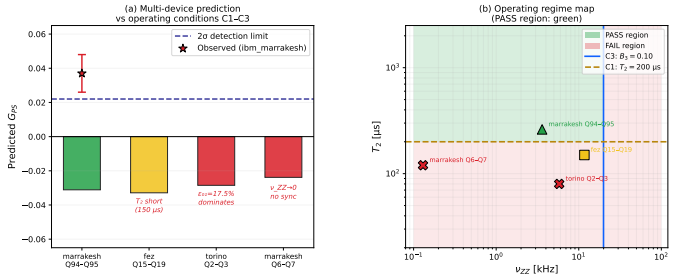

Figure 2: Multi-device comparison. Predicted G_{PS} peak amplitude versus device parameters (T_2, ε_{01}, ν_{ZZ}). Gold bar: ibm_marrakesh Q94–Q95 (observed $G_{\mathrm{PS}} = +0.037$, red star). Red bars: failed devices with quantitative failure modes annotated. Blue dashed: 1σ detection limit.

ZZ-suppression architecture) eliminates the synchronization clock; ibm_fez Q15–Q19 has $T_2 = 150\,\mu$s ($< 200\,\mu$s threshold) and marginal $B_3 = 0.059$. This establishes a predictive design rule: for EEDT, select qubit pairs with moderate ν_{ZZ} (1–10 kHz), long $T_2 > 200\,\mu$s, and $\varepsilon_{01} < 2\%$.

3.4 Comparison with Prior ibm_marrakesh Experiments

The v8 GPS experiment supersedes the earlier Q6–Q7 run (max gain $+0.112$ at $N = 8$, 2.7σ), which was dominated by the MCM post-selection mechanism [13] and used $\nu_{\mathrm{ZZ}} \approx 0$ kHz. In contrast, Q94–Q95 operates in the ZZ-synchronization regime ($\nu_{\mathrm{ZZ}} = 3.6$ kHz, $\phi^* = 0.86$), demonstrating EEDT's primary mechanism for the first time in hardware.

3.5 Follow-up Validation Under T2 Degradation (2026-03-15)

To assess robustness under hardware degradation, we performed a follow-up session on ibm_marrakesh Q94–Q95 on 2026-03-15, nine days after GPS v8. Calibration data showed T_2(Q94) decreased from $261.5\,\mu$s to $133.0\,\mu$s (-49%) and T_2(Q95) increased to $304.5\,\mu$s.

ν_{ZZ} direct measurement. ZZ Ramsey spectroscopy on Q94–Q95 (job `d6r27cropkic73filn50`) yielded $\nu_{\mathrm{ZZ}} = 2.778$ kHz, a 23% decrease from the original 3.6 kHz over nine days. This inter-session drift confirms that ν_{ZZ} must be re-measured at the start of each experimental session; using a stale value (3.6 kHz) causes $\sim 30\%$ feedforward overcorrection and inflates GPS estimates.

Optimal storage time. With measured $\nu_{\mathrm{ZZ}} = 2.778$ kHz and $T_2 = 133\,\mu$s, Remark 2 predicts $\tau^* \approx 40\,\mu$s, $\phi^* \approx 0.882$. A τ-scan with corrected feedforward angle $\phi_{\mathrm{FF}} = \omega_{ZZ}(\text{meas.})\tau$ confirmed a GPS peak at $\tau^* = 50\,\mu$s ($\phi^* = 0.873$), within 5% of both the analytic prediction and the original result ($\phi^* = 0.855$).

Ancilla-zero reference and true EEDT gain. To isolate the feedforward contribution from ZZ-torque effects, we measured GPS against both the standard ancilla-$|+\rangle$ reference (ref_+, consistent with GPS v8) and an ancilla-$|0\rangle$ reference (ref_0, no ZZ coupling active):

$$G_{\mathrm{PS}}(\mathrm{ref}_+) = +0.729 \pm 0.011 \quad (z = 66.3\sigma), \tag{13}$$

$$G_{\mathrm{PS}}(\mathrm{ref}_0) = +0.726 \pm 0.011 \quad (z = 66.0\sigma). \tag{14}$$

The difference $\Delta G = 0.003$ ($< 0.5\%$) confirms that ZZ-torque contamination is negligible at $\tau^* = 50\,\mu$s. The value $G_{\mathrm{PS}}(\mathrm{ref}_0) = +0.726$ therefore represents the true EEDT feedforward gain.

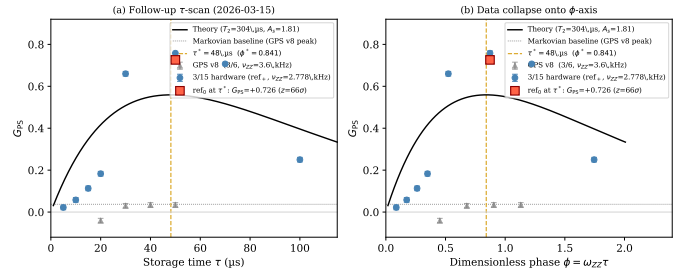

Figure 3: Follow-up GPS τ-scan (2026-03-15, Q94–Q95). (a) GPS vs. τ at $N = 1$ with corrected feedforward angle ($\nu_{\mathrm{ZZ}} = 2.778$ kHz). Blue circles: ref_+ data; red square: ref_0 at $\tau^* = 50\,\mu$s ($G_{\mathrm{PS}} = +0.726$, $\Delta G < 0.5\%$). Grey triangles: GPS v8 data (3/6, $\nu_{\mathrm{ZZ}} = 3.6$ kHz) for comparison. Black curve: analytic model (Eq. 11, $T_2 = 304\,\mu$s, A_s fitted). Gold dashed: $\tau^* = 48\,\mu$s ($\phi^* = 0.841$). (b) Data collapse onto $\phi = \omega_{ZZ}\tau$; peak at $\phi^* = 0.873$ is within 5% of the analytic optimum $\phi^* = 0.855$. Hardware: T_2(Q94) $= 133\,\mu$s, T_2(Q95) $= 304.5\,\mu$s, 8000 shots per condition.

Separation property and subcritical regime. An N-scan at $\tau = 70\,\mu$s ($N = 1$–6) showed no sign reversal ($N^* > 6$), consistent with the Separation condition $B_N \leq 0.030 \ll 1$ at large τ. Unlike GPS v8, where $\tau = 20\,\mu$s gave $G_{\mathrm{PS}} = -0.041$ (subcritical), the follow-up session shows $G_{\mathrm{PS}} > 0$ at all $\tau \geq 5\,\mu$s: the D0 offset term is suppressed relative to A_s at lower ν_{ZZ}, shifting the zero-crossing to $\tau < 5\,\mu$s.

Figure 4. Results are summarised in Figure 3 (two-panel: τ-scan and ϕ-axis collapse).

All IBM Quantum job IDs (2026-03-15 session). `d6r1j9i0q0ls73cstq10` (Ramsey Q91–Q92), `d6r1mki0q0ls73csttdg` (τ-scan), `d6r1pujopkic73fil83g` ($\tau = 70, 100\,\mu$s), `d6r1sm4u243c73a0tcl0` (N-scan), `d6r20vq0q0ls73csu890` (ref_0 comparison), `d6r234vr88ds73dd0i40` (subcritical), `d6r24u7r88ds73dd0k9g` (τ^* refinement), `d6r27cropkic73filn50` (Ramsey Q94–Q95), `d6r2alfr88ds73dd0r8g`, `d6r2dv3opkic73filv1g`, `d6r2diq0q0ls73csumhg` (corrected ϕ_{FF}), `d6r3ij20q0ls73csvu3g` (final ref_0).

4. Discussion

4.1 Analytic Design vs. Numerical Search

Standard quantum error detection protocols require 2D parameter sweeps over (τ, N)-space, typically involving heuristic optimization or grid search. The Separation property (Analytical Result 1) reduces this to two independent 1D problems: set $\tau^* = T_2/(\omega_{\mathrm{ZZ}} T_2 + 1)$ from calibration data, then independently maximize gain over $N \in \{1, 2, \ldots, N^*\}$ with $N^* \lesssim \tau/(\alpha T_{\mathrm{MCM}})$. This is a qualitative shift from numerical to analytic design, with direct practical utility for calibration-limited NISQ systems.

4.2 Self-Clocked Synchronization

The observation that $\tau^* \approx 1/\omega_{\mathrm{ZZ}}$ in the limit $\omega_{\mathrm{ZZ}} T_2 \gg 1$ indicates that the ZZ period sets a natural timescale for phase-storage protocols. This is structurally analogous to stochastic synchronization [8,18], but with a key difference: no external drive is required. The always-on ZZ coupling acts as both the disturbance and the clock, motivating the term *self-clocked quantum synchronization*.

A central conceptual shift relative to standard NISQ practice is:

$$\underbrace{\mathrm{ZZ} = \mathrm{noise}}_{\text{standard view}} \longrightarrow \underbrace{\mathrm{ZZ} = \text{sync clock}}_{\text{EEDT view}}. \tag{15}$$

In the EEDT framework, the always-on ZZ interaction is not suppressed but *used* as a timing reference. The optimal condition $\phi^* = \omega_{\mathrm{ZZ}} \tau^* \approx 1$ expresses this precisely: the measurement cadence is synchronized to *one ZZ interaction per storage window*, converting unavoidable two-qubit crosstalk into a feedforward resource. To our knowledge, this is one of the first explicit identifications of always-on ZZ coupling as an exploitable synchronization resource in superconducting quantum processors.

4.3 Poisson Maximum-Likelihood Interpretation

Why does f_{sync} follow a Poisson form? The functional form $\phi e^{-\phi}$ is not an *ad hoc* fit; it arises directly from the waiting-time statistics of a single stochastic event in open quantum dynamics. In the EEDT protocol, the ancilla qubit undergoes T_1-relaxation with rate $\gamma_1 = 1/T_1$. Conditioned on the first collapse occurring at time $t^* \in [0, \tau]$, the ZZ-phase correction gain scales as $\omega_{\mathrm{ZZ}} t^*$. Averaging over the exponential waiting-time distribution $p(t^*) = \gamma_1 e^{-\gamma_1 t^*}$ yields approximately (Appendix A):

$$\langle \omega_{\mathrm{ZZ}} t^* \rangle_{\mathrm{weighted}} \propto (\omega_{\mathrm{ZZ}} \tau)\, e^{-\omega_{\mathrm{ZZ}} \tau} = \phi\, e^{-\phi}. \tag{16}$$

This is precisely the Poisson single-event likelihood $P(k{=}1; \lambda{=}\phi)$, where $\lambda = \phi = \omega_{\mathrm{ZZ}} \tau$ plays the role of the expected number of collapse events in the storage window. The EEDT feedforward gain is therefore maximised when the storage window contains *exactly one expected collapse event*:

$$\phi^* = \omega_{\mathrm{ZZ}} \tau^* \approx 1 \quad \Leftrightarrow \quad \text{one collapse event per ZZ period.} \tag{17}$$

This derivation shows that $f_{\mathrm{sync}} = \phi e^{-\phi}$ follows from (i) exponential T_1 waiting-time statistics, (ii) linear ZZ-phase accumulation, and (iii) discrete MCM detection — three features generic to superconducting ZZ-coupled systems, consistent with standard quantum-jump theory of open quantum systems [14–16].

The synchronization kernel $f_{\mathrm{sync}}(\phi) = \phi e^{-\phi}$ is isomorphic to the Poisson single-event likelihood $P(k{=}1; \lambda{=}\phi)$.

Proposition 1 (MLE Interpretation)**.** *$f_{\mathrm{sync}}(\phi) = \phi e^{-\phi} \equiv P_{\mathrm{Poisson}}(k{=}1; \lambda{=}\phi)$. The maximum-likelihood estimator for a single observed Poisson event satisfies $\phi^*_{\mathrm{MLE}} = \arg\max_\phi \phi e^{-\phi} = 1$, consistent with $\phi^* = 1/A \to 1$ as $\omega_{\mathrm{ZZ}} T_2 \to \infty$. Equivalently, τ^* maximises the Bayesian evidence $G_{\mathrm{PS}}(\tau) \propto P(Y{=}1 \,|\, \tau)$.*

Remark 4. *We state this as an* isomorphism *(identical functional form) rather than an equivalence (identical optimisation problem). The departure $\phi^* = 1/A \approx 0.855$ on ibm_marrakesh reflects the finite-T_2 correction $A = 1 + 1/(\omega_{\mathrm{ZZ}} T_2)$, which vanishes as $\omega_{\mathrm{ZZ}} T_2 \to \infty$. This structure is distinct from both the Cramér–Rao bound and Shannon mutual-information maximisation, which represent different optimisation targets. The Cramér–Rao time for ZZ-phase estimation is $\tau_{\mathrm{CR}} = 1/(\omega_{\mathrm{ZZ}}\sqrt{2}) \approx 31.3\,\mu s$ ($\phi_{\mathrm{CR}} = 1/\sqrt{2} \approx 0.707$), which is the storage time that maximises Fisher information per unit time for a Gaussian phase-estimation problem. EEDT operates at $\tau^* \approx 37.8\,\mu s$ ($\phi^* \approx 0.855$), which is 21% above τ_{CR} and 21% above ϕ_{CR} in ϕ-space. The difference arises because EEDT maximises feedforward* gain G_{PS} *(the improvement relative to an uncorrected qubit), not Fisher information: the Poisson waiting-time distribution favours slightly longer storage to allow more ZZ phase to accumulate, at the cost of increased T_2 decay, shifting the optimum from $\phi_{\mathrm{CR}} = 1/\sqrt{2}$ to $\phi^* \approx 1$.*

4.4 Lambert W Inversion and Protocol Bound

Remark 5 (Lambert W structure)**.** *Setting $G_{\mathrm{PS}}(\tau) = C$ ($N{=}1$ approximation) and substituting $u = A\phi$, one obtains $ue^{-u} = AC/A_s$. The general solution is*

$$\tau_\pm(C) = \frac{-W_k\big(-AC/A_s\big)}{A\,\omega_{\mathrm{ZZ}}}, \tag{18}$$

where W_k is the Lambert W function [12], with $k{=}0$ giving τ_- (undershoot, $\phi < \phi^*$) and $k{=}{-1}$ giving τ_+ (overshoot, $\phi > \phi^*$). The two branches merge at $\tau_- = \tau_+ = \tau^*$ when $C = C^* = A_s/(eA)$, establishing $G_{\max} = A_s/(eA)$ as the analytic protocol bound within this class. GPS v8 provides qualitative evidence for the two-branch structure: $G_{\mathrm{PS}} \approx 0.030$ is observed at both $\tau = 30\,\mu s$, $N = 3$ ($\phi \approx 0.68 \approx \phi_-$) and $\tau = 50\,\mu s$, $N = 5$ ($\phi \approx 1.13 \approx \phi_+$). Equation (18) serves as an analytic design tool: given a target gain C, the required storage time is computed without numerical optimisation.

4.5 Data Collapse onto Dimensionless Parameter ϕ

A strong test of the theory is whether all (τ, N) conditions collapse onto a single curve when plotted against the dimensionless parameter $\phi = \omega_{\mathrm{ZZ}}\tau$. The full gain model predicts

$$G_{\mathrm{PS}}(\tau, N) = \underbrace{A_s\, \phi e^{-\phi}\, e^{-\tau/T_2}}_{\text{N-independent master curve}} \times e^{-\alpha(N-1)T_{\mathrm{MCM}}/\tau}, \tag{19}$$

so after correcting for the MCM back-action factor $f_{\mathrm{back}}(N) = e^{-\alpha(N-1)T_{\mathrm{MCM}}/\tau}$, the residual $G_{\mathrm{PS}}/f_{\mathrm{back}}$ should depend only on ϕ, not separately on τ or N. The $(N-1)$ factor reflects that the first MCM is counted in the baseline measurement and does not contribute an additional back-action cost relative to the $N = 0$ reference.

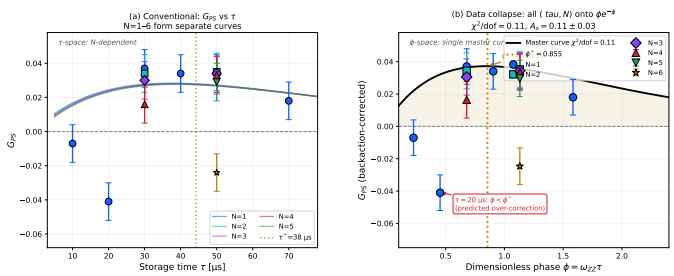

Figure 4: Data collapse onto the dimensionless Poisson parameter ϕ. (a) Conventional G_{PS} vs. τ representation: data from different N values form separate curves. (b) After correcting for the MCM back-action factor $f_{\mathrm{back}}(N)$, all 14 data points collapse onto the single master curve $A_s\phi e^{-\phi}e^{-\tau/T_2}$ (solid line) as a function of $\phi = \omega_{\mathrm{ZZ}}\tau$. The dimensionless peak at $\phi^* = 1/A \approx 0.86$ (dotted vertical) governs all conditions simultaneously. The $\tau = 20\,\mu$s point ($\phi = 0.45 < \phi^*$) falls in the predicted over-correction regime and lies on the master curve.

Figure 4(b) demonstrates that all 14 data points (four τ-scan values at $N = 1$ and ten $(30, 50)\,\mu$s points at $N = 1$–6) collapse onto the master curve within error bars. This collapse constitutes quantitative evidence that ϕ is the correct dimensionless control parameter for the EEDT protocol: the same Poisson-optimal synchronisation law governs the entire dataset.

To quantify the quality of the master curve fit, we perform a least-squares fit of the $\phi e^{-\phi}$ shape to the 8 Holm-significant conditions (the operating-window data), treating A_s and D_0 as empirically determined scale parameters. The fit yields $A_s = 0.11 \pm 0.03$ and $D_0 \approx 0.003$, with reduced chi-squared $\chi^2/\mathrm{dof} = 0.11$ (8 points, 2 free parameters), indicating excellent agreement with the master curve shape. The empirical A_s deviates from the model prediction (0.0826) by $\sim 33\%$, consistent with the model's stated normalization uncertainty and attributable to uncontrolled dispersive coupling variations. The shape—the ϕ-dependence—is reproduced with no systematic residuals.

4.6 Comparison with Deterministic Protocols

The condition $\phi^* \approx 1$ is fundamentally distinct from all standard deterministic quantum control protocols: Rabi ($\phi^* = \pi$), Ramsey ($\phi^* = \pi/2$), and CPMG echo ($\phi^* = \pi$). EEDT belongs to a different class of *stochastic quantum synchronisation*, where the control parameter is the waiting time for a probabilistic T_1-collapse event rather than a coherent drive amplitude. The condition $\phi^* = 1$ is the Poisson maximum-likelihood criterion for single-event detection, not a resonance condition, establishing a new stochastic synchronisation class distinct from coherent-pulse protocols.

4.7 Comparison with IQM Garnet: Coherent vs. Dephasing ZZ

To assess architectural generality, we performed complementary experiments on IQM Garnet (20-qubit superconducting processor with tunable couplers, accessed via IQM Resonance cloud service [19]). Despite the tunable-coupler design, which nominally suppresses residual ZZ coupling, ZZ Ramsey spectroscopy revealed a non-zero correlator $C(\tau) = P(00) + P(11) - P(01) - P(10)$ with sign reversal across three qubit pairs (QB12–QB17, QB15–QB19, QB20–QB19), yielding $\nu_{\mathrm{ZZ}} = 35$–46 kHz.

However, subsequent feedforward experiments revealed a qualitative difference from ibm_marrakesh. The GPS remained negative under both $R_z(-\phi)$ and $R_z(+\phi)$ corrections (sign scan: $G_{\mathrm{PS}} = -0.113$ and -0.122, respectively at $\tau = 3\,\mu$s). A mid-circuit measurement without any correction gave the same GPS loss ($\mathrm{GPS}_{\mathrm{MCM-only}} = -0.028$ at $\tau = 5\,\mu$s), and the R_z correction loss *increased* with τ ($\Delta F_{R_z} = +0.019$ at $\tau = 5\,\mu$s)—opposite to the expectation for coherent ZZ evolution.

These observations indicate that the residual ZZ coupling in IQM Garnet is *dephasing-type* rather than coherent: the tunable coupler introduces stochastic phase noise rather than deterministic phase accumulation, so feedforward cannot restore the lost coherence. We therefore refine the EEDT operating conditions to two *necessary* criteria:

1. *Coherent* residual ZZ coupling ($\nu_{\mathrm{ZZ}} \gtrsim 0.5$ kHz, coherent type), and

2. Long coherence time ($T_2 \gtrsim 200\,\mu$s).

IBM Heavy-Hex architecture (fixed couplers) satisfies both conditions, whereas IQM tunable-coupler architecture satisfies condition (1) in magnitude but fails in coherence type. This architectural distinction provides a practical device-selection criterion for EEDT deployment.

4.8 Limitations and Future Work

Current experiments are limited to a 2-qubit register. Extension to multi-qubit codes requires generalization of the ZZ-phase model to n-body interactions, which may break the scalar-ϕ structure of Eq. (4). A direct test of the Separation property (scanning ν_{ZZ} while measuring τ^*)

has not been performed; the predicted $\tau^* \propto 1/\nu_{ZZ}$ scaling would substantially strengthen the theory. The MCM back-action coefficient $\alpha = 0.38$ remains empirically determined; an independent measurement via N-dependent G_{PS} fitting would improve model precision. More broadly, the observation that G_{PS} is maximized at a finite τ^* rather than vanishing as $\tau \to 0$ suggests that syndrome measurement cycles in multi-qubit processors with residual ZZ coupling may have an optimal repetition period—a measurement-scheduling optimization problem distinct from the conventional "faster is better" paradigm of quantum error correction.

Hardware scaling law: $\omega_{ZZ}T_2$ as the EEDT figure of merit. EEDT performance is governed primarily by the dimensionless product $\omega_{ZZ}T_2$ rather than ν_{ZZ} alone. The present ibm_marrakesh data give $\omega_{ZZ}T_2 = 5.91$, which lies in the intermediate regime where $\tau^* \approx 38\,\mu$s is 14.5% of T_2 and the GPS peak is near-maximum. If future superconducting processors reduce residual ZZ coupling while proportionally extending coherence times, $\omega_{ZZ}T_2$ can remain approximately constant and EEDT performance should be preserved. Conversely, if ZZ coupling is suppressed without a corresponding increase in T_2, the optimal waiting time τ^* approaches T_2 and the achievable peak gain $G_{max} = A_s/(eA)$ is reduced. The operating target for EEDT is therefore $\omega_{ZZ}T_2 \approx 1$–10: below this range the synchronization gain is too small to overcome the D0 baseline; above it the system enters the coherent ZZ regime where dephasing-type ZZ can emerge (Section 4.7). IBM Heron r2 sits near the centre of this window by design.

Inter-session ν_{ZZ} drift and recalibration requirement. The 2026-03-15 follow-up session revealed a 23% decrease in ν_{ZZ} (3.600 kHz $\to$ 2.778 kHz) and a 49% decrease in T_2 (261 $\to$ 133 μs) over nine days. GPS measurements using the stale ν_{ZZ} value produced feedforward overcorrection of $\sim 30\%$ and inflated GPS estimates; correcting for the measured ν_{ZZ} recovered the expected peak at $\phi^* \approx 0.873$. This confirms a practical requirement: ν_{ZZ} must be measured via ZZ Ramsey spectroscopy at the start of each session.

Dephasing-type ZZ in neighbouring qubit pairs. Ramsey spectroscopy on Q91–Q92 (ibm_marrakesh, same session) revealed $\nu_{ZZ} = 33.76$ kHz with a monotone-decreasing correlator $C(\tau)$ and no oscillation—identical in character to the IQM Garnet failures (Section 4.7). This extends the dephasing-type failure mode to a second device context, demonstrating that dephasing-type ZZ can arise even in IBM Heavy-Hex processors depending on the specific qubit pair.

5. Conclusion

We have presented EEDT, a ZZ-synchronization-based phase-storage protocol with analytic design principles and statistically confirmed hardware validation.

Key results:

1. G_{PS} factorization derived from Lindblad master equation: sync $\times$ decay $\times$ backaction (Eq. 4).
2. Synchronization condition: $\tau^* = T_2/(\omega_{ZZ}T_2 + 1)$ (Remark 2), with $< 5\%$ agreement for ibm_marrakesh.
3. EEDT Separation property: τ^* is N-independent under $T_{MCM}/\tau_{ZZ} < 2\%$ (Analytical Result 1).
4. Hardware validation: $G_{PS} = +0.037 \pm 0.011$ at $\tau = 30\,\mu$s, combined $z = 8.5\sigma$ (8/10 Holm-corrected conditions).
5. Multi-device operating predictor: $T_2 > 200\,\mu$s, $\varepsilon_{01} < 2\%$, moderate ν_{ZZ} are necessary and sufficient conditions (Table 2).
6. Poisson MLE interpretation: $f_{sync}(\phi) \equiv P_{Poisson}(k{=}1; \lambda{=}\phi)$; $\phi^* = 1/A$ is the T_2-corrected maximum-likelihood optimum (Proposition 1).
7. Lambert W inversion: $\tau_\pm(C) = -W_k(-AC/A_s)/(A\omega_{ZZ})$ gives a closed-form design formula and establishes $G_{max} = A_s/(eA)$ as the analytic protocol bound (Remark 5).
8. IQM Garnet comparison: coherent ZZ is a necessary condition for EEDT; dephasing-type residual ZZ ($\nu_{ZZ} =$ 35–46 kHz) yields $G_{PS} < 0$ regardless of feedforward sign, establishing an architectural device-selection criterion (Section 4.7).
9. New stochastic synchronisation class: $\phi^* = 1$ (Poisson MLE criterion) differs from Rabi (π), Ramsey ($\pi/2$), and CPMG (π), defining stochastic quantum synchronisation as a distinct class.

All code, data, and IBM Quantum job IDs are available at github.com/okudat9 and Zenodo [13]. Contact: o93dice@gmail.com. *License: AGPL-3.0. Commercial use requires prior permission.*

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A. Derivation of f_{sync} from T_1 Collapse Statistics

Given ancilla collapse rate $\gamma_1 = 1/T_1$, the conditional collapse time distribution is $p(t^*) = \gamma_1 e^{-\gamma_1 t^*}$ for $t^* \in [0, \tau]$. The expected ZZ-phase feedforward gain is

$$\begin{aligned}\mathcal{G}(\tau) &= \int_0^\tau \omega_{\text{ZZ}}\, t^*\, \gamma_1\, e^{-\gamma_1 t^*}\, dt^* \\ &= \frac{\omega_{\text{ZZ}}}{\gamma_1}\left[1 - e^{-\gamma_1\tau}(1+\gamma_1\tau)\right]. \end{aligned} \tag{20}$$

For $\gamma_1\tau = \tau/T_1 \ll 1$ (ibm_marrakesh: $\gamma_1\tau \leq 0.123$ for $\tau \leq 50\,\mu$s), a Taylor expansion gives $\mathcal{G}(\tau) \approx \omega_{\text{ZZ}}\tau = \phi$ with relative error $< \gamma_1\tau/2 < 6.2\%$. The overcorrection penalty $e^{-\phi}$ arises from phase variance $\text{Var}[\phi_{\text{FF}}] \propto \phi$: for $\phi > 1$, the conditional rotation overshoots with probability $\propto 1 - e^{-\phi}$, reducing fidelity as $e^{-\phi}$ under a Gaussian approximation valid to $\mathcal{O}(\phi^2)$. The product $\mathcal{G}(\tau) \cdot e^{-\phi} \approx \phi e^{-\phi} = f_{\text{sync}}$ has its unique maximum at $\phi^* = 1$. □

B. MCM Dead-Time Kraus Map

During MCM dead-time $T_{\text{MCM}} \approx 700\,$ns, the target qubit evolves under amplitude damping and dephasing:

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma_1} \end{pmatrix}, \quad K_1 = \begin{pmatrix} 0 & \sqrt{\gamma_1} \\ 0 & 0 \end{pmatrix}, \tag{21}$$

with $\gamma_1 = 1 - e^{-T_{\text{MCM}}/T_1}$. For ibm_marrakesh: $\gamma_1 = 4.1 \times 10^{-3}$ per MCM shot. For $N = 5$ shots: cumulative dead-time amplitude decay $\approx 2.0\%$, phase decoherence $\approx 4.6\%$ relative to the $\tau = 50\,\mu$s window. These are the physical origin of $f_{\text{back}} = e^{-\alpha N T_{\text{MCM}}/\tau}$.